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A new design for a versatile Fabry-Pérot interferometer for Brillouin spectroscopy

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We present a new design of a Fabry-Pérot interferometer for Brillouin spectroscopy. Emphasis is placed on a quick change of the free spectral range without losing alignment, the possibility of switching the multipass arrangement from 3 to 5 pass and back, and finally on a simple and inexpensive construction. Besides this the new interferometer shows very good long term stability and is easy to handle. These features were not available before in this combination. The performance of the instrument will be demonstrated. © 1997 American Institute of Physics. [S0034-6748(97)04811-9]

I. INTRODUCTION

The interferometer developed in 1897 by Fabry and Pérot¹ is used in many areas of modern science. In high resolution Brillouin spectroscopy, Fabry-Pérot interferometers (FPIs) are widely used since they possess a high resolving power and contrast. Over many years it turned out that plane piezoelectric scanned FPIs are the most suitable instruments for this purpose.

Our aim was to design an inexpensive FPI optimized for Brillouin spectroscopy of strong Rayleigh scatterers like polymers. This needs high contrast, moderate finesse, and long term stability.

The most common design uses a symmetric structure based upon three low expansion rods which are connected to end plates. One of the mirrors is attached to one of these plates. Approximate alignment is achieved by means of differential screws which tilt that mirror. The other mirror is attached through three piezoelectric scanning stacks to a sledge which can be moved along the rods to change the mirror spacing. The exact parallel alignment of the mirrors is achieved by applying suitable bias voltages to the three piezo stacks. The instrument is scanned by applying a voltage ramp to all scanning piezos. Some designs use two sets of piezos, three for alignment of the mirrors and three to scan the instrument, respectively.

The main advantage of the rigid three rod design is a very good long term stability. The disadvantages are various. We will list a few below.

In general, piezoelectric transducers are nonlinear. Therefore it is hard to produce a large scan while maintaining a sufficient parallelism of the mirrors.

The device is not suitable for tandem operation. It is nearly impossible to achieve the necessary synchronous operation of two interferometers.

The use of materials with low thermal expansion increases the cost of such an instrument. Besides, these materials (e.g., Invar[®]) are very hard to machine, a point which plays an important role on homemade interferometers.

The main drawback is the way one has to change the mirror spacing. This distance determines the resolution of the instrument, and, therefore, in case the operator wants to change the resolution one has to change the mirror spacing.

More often, this procedure is necessary because the Brillouin lines from different orders start overlapping while changing, e.g., the temperature of the sample. To change the distance between the mirrors, one has to slide the scanning mirror assembly along the three rods. This leads to a loss of mirror alignment and the handling produces local temperature changes. It is very time consuming to realign the interferometer and to determine its new free spectral range (FSR) accurately. Also, time must be allowed for the instrument to regain thermal equilibrium. The whole process is very inconvenient and time consuming. Therefore, measurements requiring several changes of the mirror spacing are nearly impossible.

The other design routinely used nowadays is the Sandercock-type.² The scanning mirror is attached to a deformable parallelogram which is driven by one scanning piezo. The mirror spacing is monitored by a capacitive displacement transducer. This assembly is attached to a common translation stage for large displacements. The second mirror can be aligned by means of two piezotransducers (fine alignment) and two micrometer screws (coarse alignment), respectively.

The main disadvantage is the great sensitivity of the interferometer to vibrations because of the tuning forklike design which often requires the use of an active antivibration system. The advantages of the Sandercock design are good overall performance and a simple way to extend the design for tandem operation.³ One of the major problems of a FPI, already mentioned above, is the overlapping of Brillouin lines of different orders. This can be solved either by an appropriate changing of the free spectral range or by using a tandem arrangement.

The drawbacks of a tandem interferometer are well known: the difficult alignment, the great sensitivity of the whole system against vibration, the problem of maintaining a constant ratio of the mirror spacings over the whole scan range (not with the Sandercock design) and, in case one wants to observe weak features in the spectrum, the sometimes insufficient suppression of the ghosts. The Sandercock-type of a tandem interferometer is known to overcome most of these drawbacks, but the instrument commercially available (JRS Scientific Instruments) is expensive compared to the cost we aspired to achieve. We decided to choose another

way to solve the problem of overlapping lines and to build a new type of Fabry-Pérot interferometer, which allows the free spectral range to be changed without losing the alignment of the interferometer and also meets all our other demands like long term stability, low sensitivity against vibration, high transmitted intensity, and high contrast.

II. THEORY OF OPERATION

In an ideal case (reflecting surfaces perfectly flat and diffraction neglectable) the transmission function for a plane FPI is given by an Airy function:

$$I(\delta) = I_0 \left(1 - \frac{A}{(1-R)} \right) \frac{1}{1 + C \sin^2 \delta/2}, \quad (1)$$

where A is the absorption coefficient of the mirror coating, R the reflectivity, and C the contrast given by

$$C = \frac{I_{\max} - I_{\min}}{I_{\min}} = \frac{4R}{(1-R)^2}. \quad (2)$$

Changing δ leads to a periodic change of the intensity. Since δ is given by

$$\delta = \frac{4\pi n d \cos \alpha}{\lambda_0} \quad (3)$$

one can do this either by changing the index of refraction n of the medium between the mirrors, the mirror spacing d , or the incidence angle α (λ_0 being the vacuum wavelength of the incident light).

The most commonly used way in modern Brillouin spectroscopy is to change the mirror spacing by piezoelectric scanning stacks, using perpendicular incidence of the light ($\alpha = 0^\circ$) and working at ambient condition in air ($n \approx 1$).

In this case the FSR (which is the separation between two consecutive maxima of the Airy function) is given by

$$\text{FSR} = \frac{c}{2d} \quad (4)$$

(c speed of light).

The finesse of a FPI, which is defined as the ratio of the FSR to the half-width of the instrumental function, is

$$F = \pi \frac{\sqrt{R}}{(1-R)}. \quad (5)$$

The finesse determines the resolving power of a FPI.

Deviations from the ideal case as described here are discussed widely in literature (e.g., Refs. 4–8), but do not alter the principle behavior.

Brillouin spectroscopy normally requires a higher contrast than achievable through passing the interferometer once. This requirement can be fulfilled by multipass systems. The contrast for p passes is given by

$$C_p = \left(\frac{4R}{(1-R)^2} \right)^p = C_1^p \quad (6)$$

and the finesse

$$F_p = (2^{1/p} - 1)^{-1/2} \pi \frac{\sqrt{R}}{(1-R)} = (2^{1/p} - 1)^{-1/2} F_1. \quad (7)$$

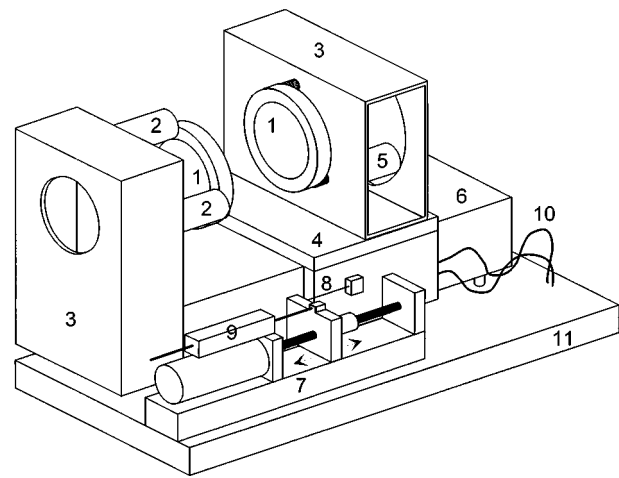


FIG. 1. A scheme of the interferometer [(1) mirrors, (2) piezo transducers, (3) mirror holders, (4) movable sledge, (5) motorized micrometer screw, (6) Zerodur® block, (7) sledge drive with motor and spindle, (8) connection between sledge and drive, (9) linear resistor, (10) air supply and vacuum hoses for the air bearings, (11) ground plate].

The possibility of increasing contrast and finesse by multipassing the interferometer is limited by the loss of intensity at each pass, which is caused mainly by absorption and surface imperfections. Another factor is a possible deviation from perfect parallelism of the two mirrors, which causes different δ for every pass. This reduces the transmission because of the imperfect overlap of the single pass transmission function.

III. DESIGN

A. Interferometer

The whole instrument is based on a Zerodur® block. We use Zerodur® because of its very small thermal expansion coefficient, which makes thermally induced change of the cavity length negligible. Besides, Zerodur® can be polished very well to achieve flat surfaces.

This is important because the block serves as a guide for the sledge on which one of the mirrors is mounted. The sledge moves on air bearings (Anaspec). This ensures together with the high quality surfaces of the Zerodur® block a tilt free movement over several centimeters necessary to change the FSR without losing alignment. The sledge is moved by a motorized spindle. After moving, the sledge is fixed through vacuum generated by a pump.

The mirror holder on the sledge holds three motorized micrometer screws. These and their control electronics are homemade and allow a remote coarse alignment with a resolution of better than $0.1 \mu\text{m}$.

The other mirror holder is attached to one end of the Zerodur® block and holds the three piezo transducers (Queensgate) with a capacitive displacement control. These are used for scanning of the interferometer and fine alignment of the mirrors.

A schematic of the interferometer is shown in Fig. 1.

Besides the Zerodur® block, the interferometer is made from aluminum. Aluminum is inexpensive, easy to machine,

and has a high thermal conductivity. Therefore, the instrument regains thermal equilibrium very fast. The disadvantage of aluminium is of course the high thermal expansion, which could influence the free spectral range during temperature changes. We minimize that by a special arrangement of the mirrors. All remaining temperature effects are compensated by the computer control.

One mirror is mounted close to the center of gravity of the whole assembly of sledge and mirror holder. Thus, any influences through thermal induced length changes are almost compensated. The other mirror holder is attached directly to one end of the Zerodur® block and the piezotransducer housing is made from Invar®, both materials with very small thermal expansion coefficient.

The mirror spacing is determined by a linear resistor (Novotek), which measures the travel of the sledge drive. Calibrated once, we achieve an accuracy of better than 5% in determining free spectral ranges normally used in Brillouin spectroscopy.

We use 2 in. mirrors with a flatness of $\lambda/200$ at 632.6 nm (Burleigh). The reflectivity is 86% at 532 nm and optimized for a high throughput with a high contrast (five pass) rather than a maximum resolving power.⁸

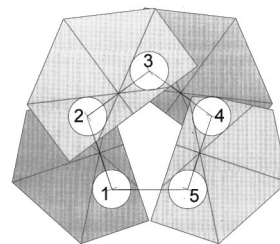
The interferometer is placed on two heavy iron plates (100 kg). The first one sits on thin rubber sheets to damp high frequency vibrations. Between the first and the second plate are three steel balls to eliminate low frequency disturbances. The whole assembly is surrounded by a closed housing to suppress stray light, noise, and temperature changes from the surroundings. Because all components of the interferometer are controlled remotely, one is able to operate the instrument without opening the housing.

B. Interferometer control

Two personal computers are used for controlling the system. One is used to run the program which scans and controls the interferometer, the other for data acquisition.

The control program includes a drift and an alignment algorithm. The drift control stabilizes the central Rayleigh line at a fixed position. First the channel with the maximum number of counts of the latest scan is taken. Around this channel, two windows, each 18 channels broad with a distance of 12 channels between them, are taken and the intensity of each window is added up. The two windows are shifted until the sum of the left window becomes greater than that of the right window. In this case, the Rayleigh peak position is the middle of the windows. If the drift (difference between starting and actual position) is greater than one channel, this spectra will not be added to the others, but the mirror spacing is corrected to have the starting position before the next scan. The alignment control keeps the mirrors parallel through optimizing the intensity. If there are five consecutive scans with the Rayleigh line intensity being smaller than the starting value, the alignment control starts an optimizing procedure: during scanning only over a few channels around the central Rayleigh line the offset of one piezo transducer is changed in one direction as long as this increases the intensity. If not, the other direction is tried.

A, 5-pass



B, 3-pass

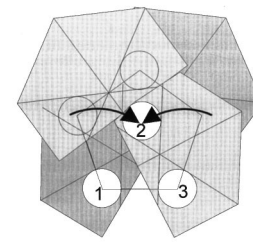


FIG. 2. Position of the corner cubes for three and five pass. The arrows indicate the rotation of the corner cubes to switch from five to three pass.

This is done with all three piezos and the obtained maximum intensity becomes the new starting value.

All parameters of the control and scan program can be altered easily to meet the demands of different measurements.

C. The 3/5-multipass option

Multipassing is the common way to enhance the contrast ratio. However, one pays with reduced luminosity. Thus the ability to change between three pass or five pass is a comforting option. In many cases, especially when investigating liquids, the three-pass option is well suited. In cases with strong Rayleigh scatterers the five-pass option is needed. There exist a number of homemade approaches for changing between single and multipass in order to attain alignment. A quick change between three pass and five pass has not come to our attention. The equipment for this task was developed by us several years ago in combination with a Burleigh FPI. It works reliably, hence, we built a second one for our new interferometer.

The basic idea is the following: the incoming and the outgoing beams are fixed in space. A change of beam folding is possible by shifting mirrors or prisms into the light path or, what we decided, to rotate corner cubes around the light beam. The principle arrangement is presented in Fig. 2.

On either side of the spectrometer a pair of corner cubes is mounted on an adjustable (x, y, Φ) holder. One of the two prisms is fixed, the other can be turned around an axis normal to the prism front face. In (A) the arrangement is shown to produce symmetric five passes. In (B) the corner cubes establishing three passes are shown. The darker ones are behind the rear mirror, the lighter ones in front of the front mirror. The light passes the spectrometer at position (1), traverses the first corner cube, and reenters the spectrometer at position (2). The further light passes follow from the numbering.

Since (1) and (5) mark the entrance and the exit optical axes the appropriate prisms can be turned around these axes. The physical axis is in the rear of the prism. The first and the last prism are rotated until their return paths coincide. This forms an unsymmetrical three pass. The prisms are rotated by a small electric motor which is switched on and off remotely. Once aligned, an interferometer running in three pass can be changed to five pass within a few seconds.

The difficulty is the first alignment of the prism. It cannot be done with the spectrometer in place. The small wedge

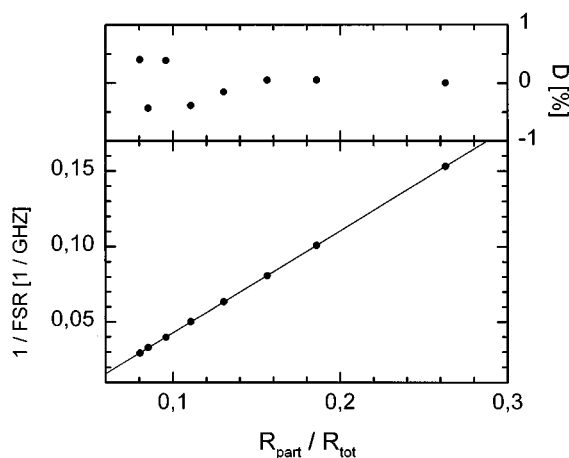


FIG. 3. Typical calibration measurements of the free spectral range with water. The top part of the plot shows the deviation D between calibration curve and measurements.

angles of the mirrors prevent this. The best method is to cut masks with the five symmetric holes using a 1:1 printout of the drawing. One mask is placed in front of the first prism pair; the second can be used at any other site. The orientation and position of the prisms are indicated on the first mask. Thus one starts pretty well. During alignment one has to minimize the vignetting of the beam at the perimeter of the throughput circle.

The rest is tedious but skillful work. Note that shifting a beam in the radial direction of a corner cub results in a symmetric response, but an angular shift results in an unsymmetric response, and the beam eventually traverses four prisms tilted with respect to each another.

The corner cubs (Spindler & Hoyer) have been ground to the final form and are anti-reflection coated. The prism housings are lined with cork. The prisms themselves are pressed against the cork by Polyamid® screws. The mountings are made from aluminum. It is fixed to the spectrometer housing.

IV. OPERATION

Crucial for a fast change of the free spectral range is a precise and convenient determination of its size. We use a linear resistor and measure the ratio between the partial (R_{part}) and total resistance (R_{tot}). The FSR can be calculated from this ratio

$$\text{FSR} = \frac{c}{2d} = \frac{\text{const}}{\frac{R_{\text{part}}}{R_{\text{tot}}} - \text{offset}}. \quad (8)$$

The constant and the offset are determined by calibration measurements with water. Figure 3 shows such a calibration.

Control measurements from time to time showed that the deviation between the FSR determined with the resistor and the measured one is smaller than 5%, a value which can be accepted. In the range normally used for Brillouin spectroscopy on polymers the deviation is much smaller.

Figure 4 shows two spectra of poly(4-methyl pentene), a

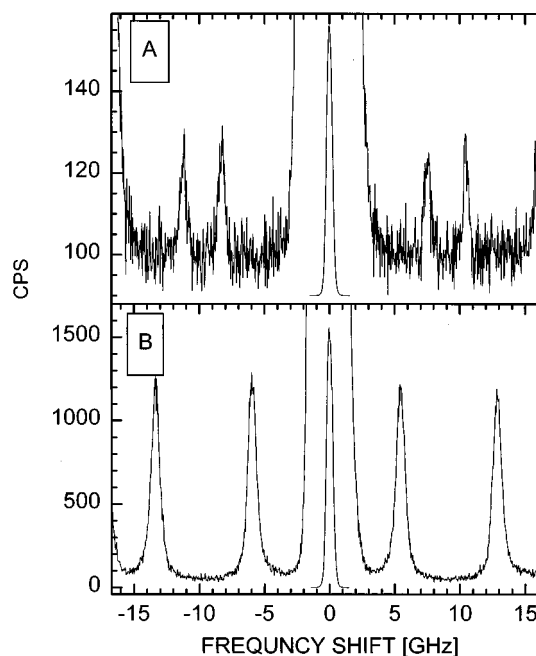


FIG. 4. Spectra of poly(4-methyl pentene-1); 5-pass, FSR 18.76 GHz, 1060 channels, dwell time/channel 1 ms. (A) at 4.9 K; 5136 accumulations (Rayleigh peak intensity is plotted reduced by a factor of 6×10^4). (B) at 308.9 K; 506 accumulations (Rayleigh peak intensity is plotted reduced by a factor of 1.5×10^3).

semicrystalline polymer, at different temperatures. A 90° scattering geometry was used⁹ and the laser power was reduced by a factor of 100 by means of a Pockels cell while scanning over the Rayleigh line (laser power was 100 mW). At 309.9 K the complete spectrum was taken within 12 min (actually only 8 min, 56 s was used for accumulation, the rest of the time was spent for flyback and scans with drift greater than one channel which were not accumulated). Even at 4.9 K only 2 h have been necessary to get an acceptable spectrum. One has to take into account that the Brillouin intensity falls linearly with temperature (a linear fit to experimental data leads to disappearing Brillouin intensity at $T = 0$ K^{10,11}).

To demonstrate the quickness of changing the free spectral range combined with a very good long term stability we made three consecutive measurements on poly(methyl methacrylate) (PMMA) (see Fig. 5). Also, the ability to change from three to five pass without time consuming alignment procedures is demonstrated. A 90° scattering geometry was used. The Brillouin lines are half-time overlapped: S_n (Stokes) and A_n (anti-Stokes) belong to the Rayleigh line R_n , the others are of different orders.

First we took a spectra with a FSR of 16.9 GHz in three-pass mode. Then we switched from three to five pass and collected data with the same FSR as before. No alignment efforts by the experimenter were necessary, only the optimizing procedure of the control program was started twice. The whole change took less than 1 min. The spectra are plotted in Fig. 5(A). The intensity of the Brillouin lines in three-pass arrangement is approximately twice as high as in five pass, but the background is lower using five passes. The finesse achieved in five pass is approximately 41.

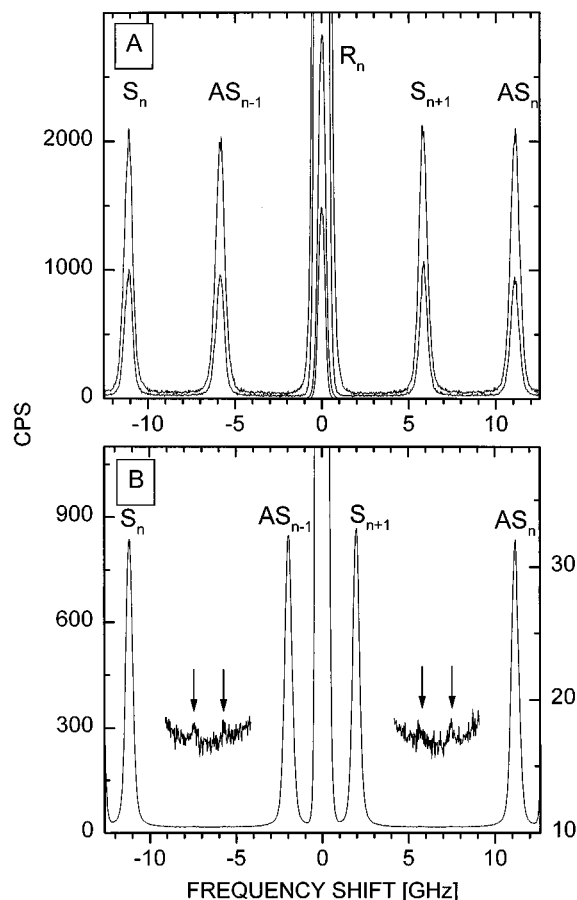


FIG. 5. Spectra of poly(methyl methacrylate) collected with different free spectral range, 1060 channels, dwell time/channel 1 ms, five pass. For assignment of the Brillouin lines see the text. (A) FSR 16.94 GHz, upper curve three pass, 1010 accumulations, lower curve five pass, 1350 accumulations (Rayleigh peak intensity is plotted reduced by a factor of 50); (B) FSR 13.14 GHz; 89.863 accumulations; the transverse Brillouin lines (arrows) are zoomed in an inset, the axis label on the right side belongs to this inset. The FSR is, for example, the separation between peaks labeled S_n and S_{n+1} .

Finally, we changed the FSR to 13.14 GHz to resolve the very weak transverse phonon of PMMA which are covered by the strong half-time overlapped longitudinal phonon at the FSR used before. The whole change of the free spectral range was performed in less than 5 min, including the optimizing procedure of the control program. Afterwards we immediately started a measurement which ran for 40 h without any intervention. This long accumulation time was necessary because the intensity of the transverse Brillouin line of PMMA is extremely weak,¹² approximately 10^4 times weaker than the longitudinal one ($I_{\text{trans}}/I_{\text{long}} = 10^{-4}$). Figure 5(B) shows the obtained spectrum. The transverse Brillouin lines are unambiguously resolved.

V. DISCUSSION

The examples from the previous section demonstrate the high performance of the instrument. Long term stability of the instrument achieved through the special design in combination with the drift and align algorithm of the computer control is evident. Our experience is that the interferometer runs stable even longer than 40 h. This means that extremely long measuring periods are possible without realigning the FPI manually. If one compares the spectrum of Fig. 4(A) to those published in the literature at the same low temperature, the advantage of a high throughput is obvious.

The possibility of switching easily and quickly between three and five pass gives the experimenter the opportunity to choose between a short collection time or a good resolution of weak features in the spectrum.

A quick change of the FSR leads to the solution of the problem of overlapping Brillouin lines.

For sure, the FPI introduced here is not a high performance instrument in the sense of an extremely high finesse but in the sense of being a highly versatile instrument for Brillouin scattering.

ACKNOWLEDGMENTS

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